

Lecture 8

Chapter 24
Electric Potential

□ Electric Potential

Imagine a charge q fixed at the origin of a coordinate system. We take another charge q_0 which we call our “**test charge**”, and we move it point **a** to point **b** under the influence of the force due to q at rest. The change in potential energy of this two-charge system is given by

$$\Delta U = \frac{qq_0}{4\pi\epsilon_0} \left[\frac{1}{r_b} - \frac{1}{r_a} \right]$$

We define the electric potential difference, ΔV to be the electric potential energy per unit test charge:

$$\Delta V = \frac{\Delta U}{q_0}; \text{ or, } V_b - V_a = \frac{U_b - U_a}{q_0}$$
$$\text{So, } \Delta V = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r_b} - \frac{1}{r_a} \right]$$

If $r_a = \infty$, and $r_b = r$, then $V_a = 0$, and $V_b = V$, represents the value of the electric potential at point **b**.

$$V(r) = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

So, the electric potential at a point is defined by the potential energy at that point per unit test charge.

$$V = \frac{U}{q_0}$$

□ Calculating the Electric Potential from the Electric Field

From the definition, the change in potential difference

$$\begin{aligned}\Delta V &= \frac{\Delta U}{q_0} \\ &= -\frac{W_{ab}}{q_0} \\ &= \frac{-\int_a^b \vec{\mathbf{F}} \cdot \overrightarrow{\mathbf{ds}}}{q_0} \\ \Delta V &= -\int_a^b \vec{\mathbf{E}} \cdot \overrightarrow{\mathbf{ds}}\end{aligned}$$

□ Electric Potential due to a charged particle

$$\vec{E} \cdot d\vec{s} = E \cos \theta ds.$$

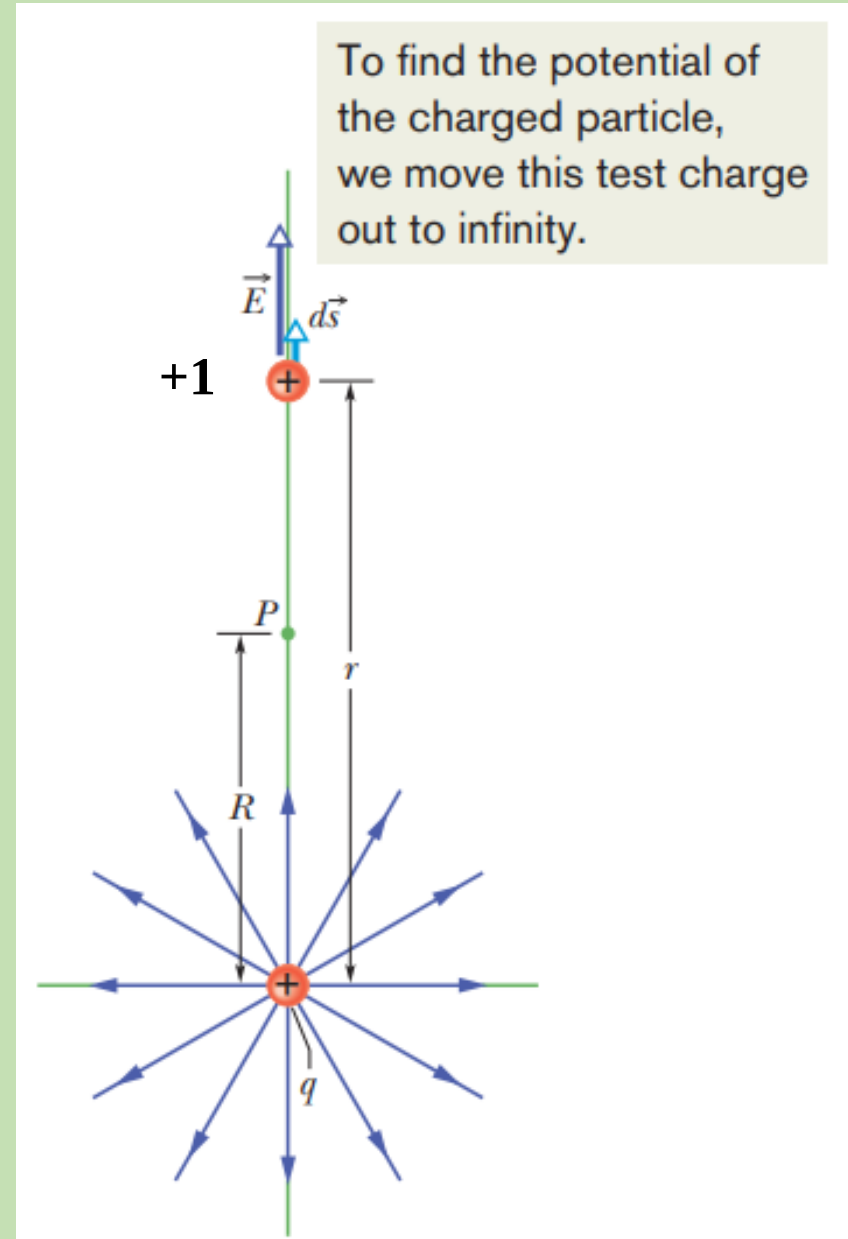
$$V_f - V_i = - \int_R^\infty E dr.$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}.$$

Next, we set $V_f = 0$ (at ∞) and $V_i = V$ (at R).

$$\begin{aligned} 0 - V &= - \frac{q}{4\pi\epsilon_0} \int_R^\infty \frac{1}{r^2} dr = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r} \right]_R^\infty \\ &= - \frac{1}{4\pi\epsilon_0} \frac{q}{R}. \end{aligned}$$

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$



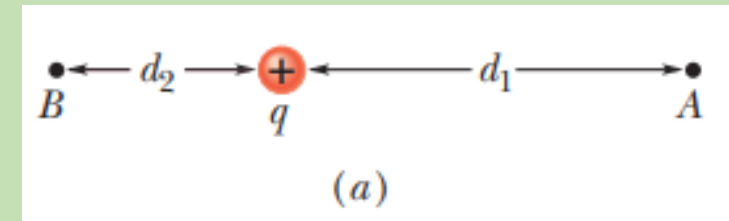
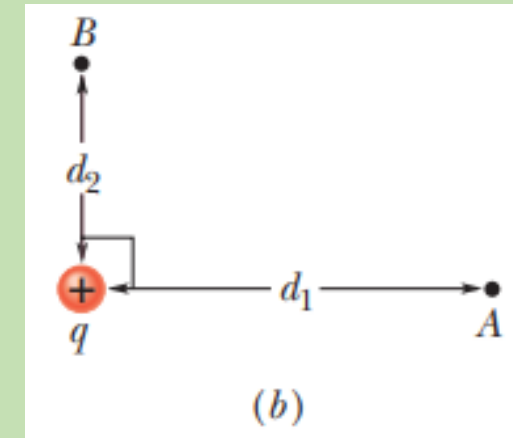
□ Potential Due to a Group of Charged Particles

The electric potential is due to a collection of charges at a particular point is defined by

$$V = \sum_{i=1}^n V_i = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i} \quad (n \text{ charged particles}).$$

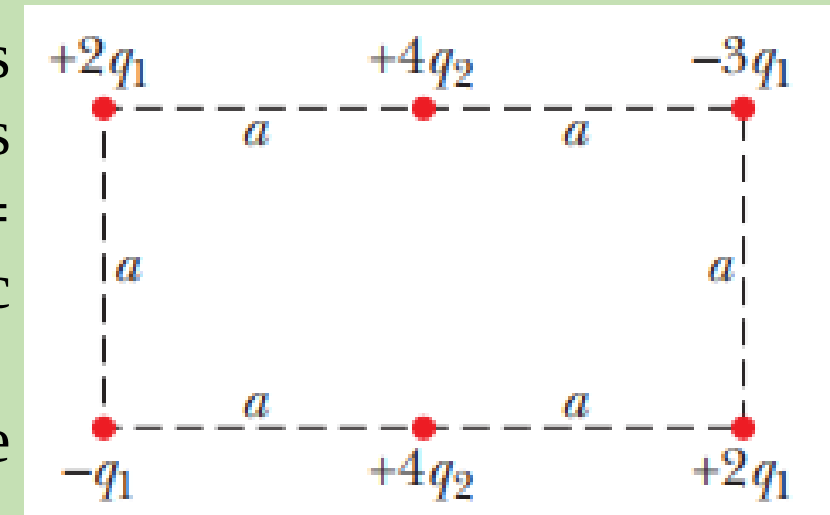
Here, q_i is the value of the i^{th} charge and r_i is the radial distance of the given point from the i^{th} charge.

Q1. Consider a particle with charge $q = 1.0 \mu\text{C}$, point A at distance $d_1 = 2.0 \text{ m}$ from q , and point B at distance $d_2 = 1.0 \text{ m}$. (a) If A and B are diametrically opposite each other, as in Figure (a), what is the electric potential difference $V_A - V_B$? (b) What is that electric potential difference if A and B are located as in Figure (b)?



Q2. Figure shows a rectangular array of charged particles fixed in place, with distance $a = 39.0 \text{ cm}$ and the charges shown as integer multiples of $q_1 = 3.40 \text{ pC}$ and $q_2 = 6.00 \text{ pC}$. With $V = 0$ at infinity, what is the net electric potential at the rectangle's center?

(Hint: Thoughtful examination of the arrangement can reduce the calculation.)



□ Potential Due to an Electric Dipole

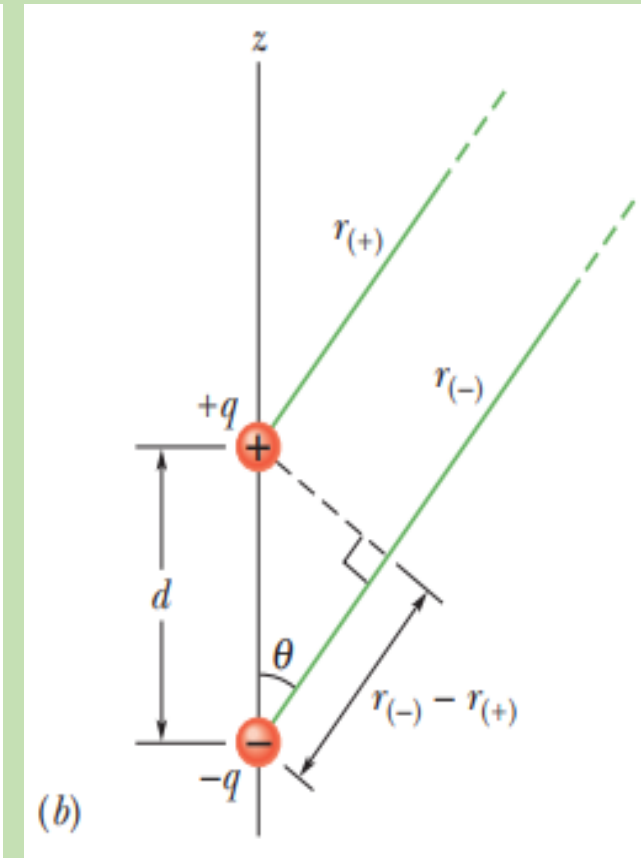
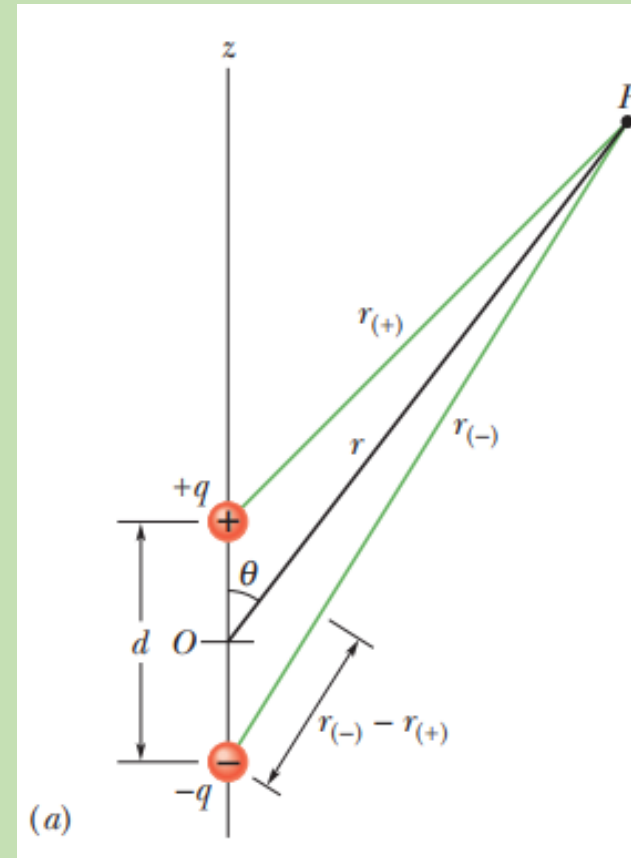
$$V = \sum_{i=1}^2 V_i = V_{(+)} + V_{(-)} = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_{(+)}} + \frac{-q}{r_{(-)}} \right)$$
$$= \frac{q}{4\pi\epsilon_0} \frac{r_{(-)} - r_{(+)}}{r_{(-)}r_{(+)}}.$$

$$r \gg d,$$

$$r_{(-)} - r_{(+)} \approx d \cos \theta \quad \text{and} \quad r_{(-)}r_{(+)} \approx r^2.$$

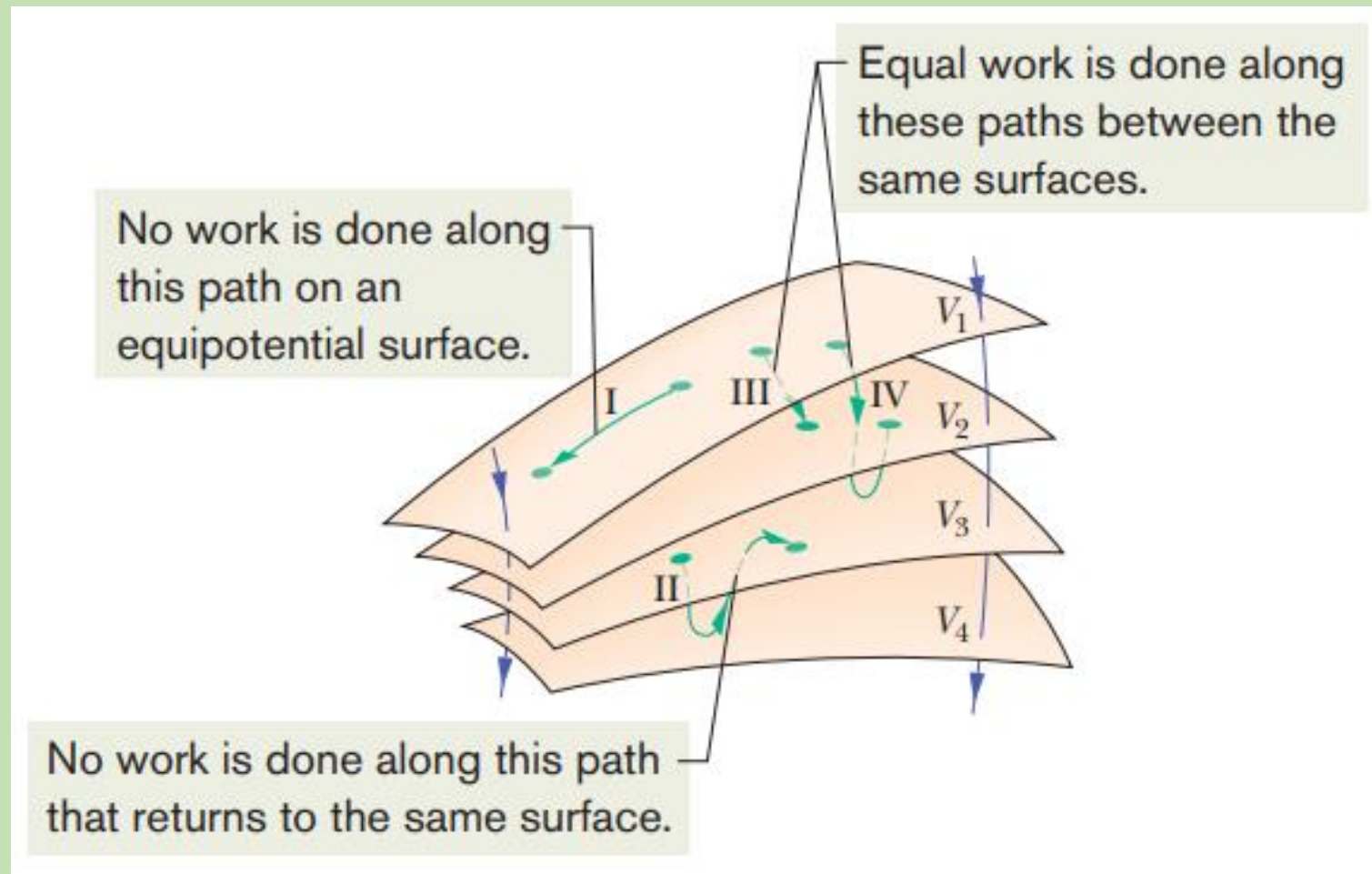
$$V = \frac{q}{4\pi\epsilon_0} \frac{d \cos \theta}{r^2},$$

$$V = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2} \quad (\text{electric dipole}),$$



Equipotential Surfaces

Adjacent points that have the same electric potential form an **equipotential surface**, which can be either an imaginary surface or a real, physical surface. No net work is done on a charged particle by an electric field when the particle moves between two points i and f on the same equipotential surface.



Q1. Two large, parallel, conducting plates are **12cm** apart and have charges of equal magnitude and opposite signs on their facing surfaces. An electric force of **$3.9 \times 10^{-15}\text{N}$** acts on an electron placed anywhere between the two plates. (Neglect fringing.) (a) Find the electric field at the position of the electron. (b) What is the potential difference between the plates?

Thank You